Md Ershadul Haque

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PROFESSIONAL SUMMARY

Mechanical engineer with expertise in finite element and discontinuous Galerkin (DG) methods for radiative transfer and electromagnetic scattering, adaptive mesh refinement, multigrid solvers, and parallel high-performance computing. Experienced in developing complete C++ simulation codes in deal.II with MPI and p4est, and in graph neural network (GNN) surrogate modeling with PyTorch Geometric. Additional industry experience in manufacturing process engineering for compressor production. Seeking industry roles in simulation, design, or R&D engineering.

EDUCATION

University of Tennessee, Knoxville, Ph.D. Mechanical & Aerospace Engineering, GPA: 3.91/4.00 — Jan 2027
King Fahd University of Petroleum and Minerals, M.S. Mechanical Engineering, GPA: 3.656/4.00 — Dec 2021
Rajshahi University of Engineering & Technology, B.S. Mechanical Engineering, GPA: 3.62/4.00 — Dec 2014

TECHNICAL SKILLS

Programming: C++, Python, MATLAB, Git
FEA/Simulation: deal.II, ANSYS, COMSOL, EES
CAD: SolidWorks, AutoCAD
Scientific Computing: HPC, MPI, p4est (parallel adaptive meshing), domain and angular decomposition
Machine Learning: PyTorch, PyTorch Geometric, graph neural networks (GNNs), physics-informed surrogate modeling
Statistics: MINITAB, regression analysis, design of experiments, parameter estimation
Modeling/Analysis: Finite element method, discontinuous Galerkin methods, radiative transfer, electromagnetic scattering, adaptive mesh refinement, multigrid methods, numerical solution of PDEs, heat transfer

RELEVANT EXPERIENCE

University of Tennessee, Knoxville — Knoxville, TN
Graduate Research & Teaching Assistant — *Jun 2022 - Present*

- Developed a space-angle discontinuous Galerkin finite element solver in C++ (deal.II) for the radiative transfer equation, including semitransparent boundary conditions and graded-index (variable refractive index) media.
- Implemented multilevel adaptive mesh refinement in both spatial and angular domains, and applied multigrid techniques to accelerate solver convergence.
- Parallelized the solver with MPI and p4est using combined angular and domain decomposition, and ran large-scale simulations on HPC clusters.
- Building GNN surrogates in PyTorch Geometric to compute scattering and phase matrices for electromagnetic scattering and coupling them into the DG radiative transfer solver (Ph.D. thesis).
- Teaching Assistant, help sessions, wrote quiz questions, and graded exams.

King Fahd University of Petroleum and Minerals — Dhahran, Saudi Arabia
Graduate Research & Teaching Assistant — *Sep 2019 - Dec 2021*

- Developed a numerical solver for the radiative transfer equation in 2D complex geometries using body-fitted (orthogonal) coordinates (M.S. thesis; published in the Journal of Non-Equilibrium Thermodynamics).
- Solved systems of PDEs in MATLAB for kinetic and reactor modeling of naphtha catalytic reforming, including literature review and model parameter estimation.
- Instructed the Materials Lab (ME 217) and graded homework and exams for mechanical engineering courses.

Walton Hi-Tech Industries Ltd., Compressor R&D — Dhaka, Bangladesh
Process Engineer — *May 2017 - Dec 2018*

- Handled product implementation and process engineering for compressor manufacturing, supporting automated lathe, milling, drilling, and grinding lines for pistons and connecting rods, plus resistance welding and washing/phosphating lines.

Mason Acoustic Company Ltd. — Dhaka, Bangladesh
Sales and Service Engineer — *Aug 2016 - Apr 2017*

- Worked toward sales targets for acoustic treatment and fireproof door products, and inspected, installed, and maintained acoustic treatment in generator rooms, auditoriums, and TV studios.

SELECTED ACHIEVEMENTS

Research output: 5 peer-reviewed journal articles (including Journal of Quantitative Spectroscopy and Radiative Transfer and IEEE Access), 1 book chapter (CRC Press), and 7 conference papers (IEEE, USNC-URSI).
Recognition: Best Paper Award, 1st International Conference on Artificial Intelligence and Data Analytics (2021); Lloyd W. Crawford Fellowship, UTSI (2022–2023); Deanship Scholarship, KFUPM (2019); Technical Board Scholarship, RUET (2011–2014).